

Transcriptome-wide Association Studies

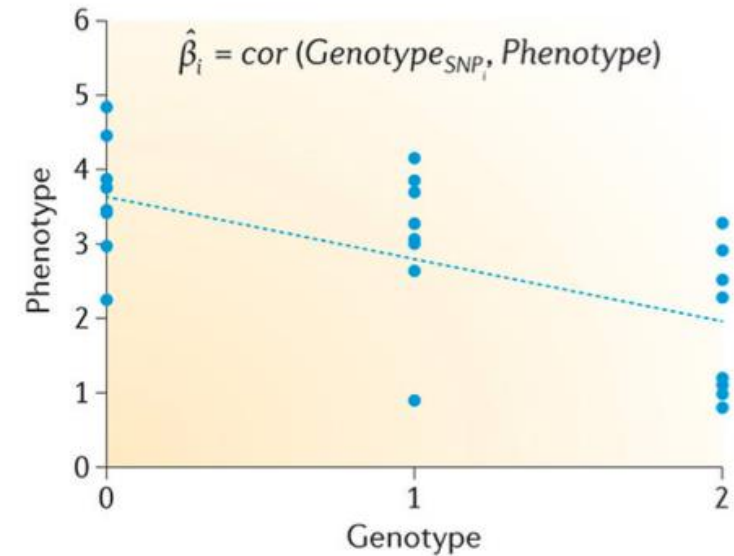
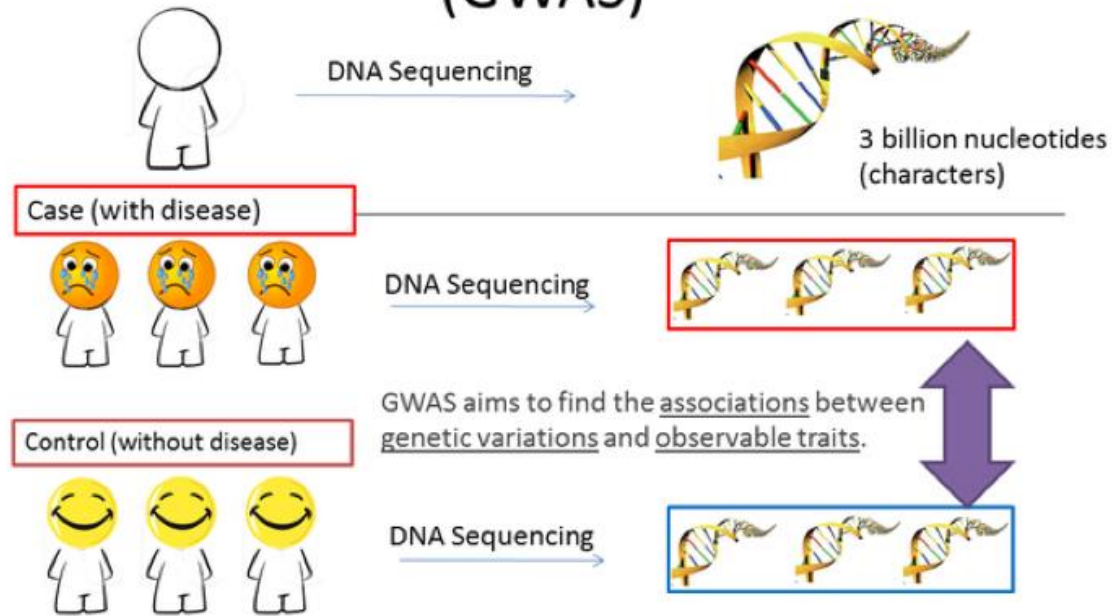
Lecture 1

Outline

- Genome-wide Association Study (GWAS)
- Transcriptome-wide Association Studies (TWAS)
 - PrediXcan/EN
 - FUSION/BestModel
 - TIGAR/DPR
 - Omnibus TWAS by ACAT

GWAS

Genome-wide Association Study (GWAS)



From [Quora.com](https://www.quora.com) and Pasaniuc B & Price AL, Nat. Rev. 2017

Study Quantitative Trait

- Linear regression model
 - $Y = \beta_0 + \alpha C + \beta_1 X + \epsilon, \epsilon \sim N(0, \sigma^2)$
 - Y represents the quantitative trait values
 - X represents the genotype data (0, 1, 2) for additive genetic model
 - C represents the confounding covariates or other environmental variables
 - ϵ represents the error term, other unknown factors
- $H_0: \beta_1 = 0 ; H_a: \beta_1 \neq 0$
- P-values can be obtained by Wald Test, T-test, Score test, Log likelihood test, etc.

Logistic Regression Model for Studying Dichotomous Phenotype

- Y = dichotomous phenotype
- X = a coding for the genotype

Genotype	Codominant	Dominant	Recessive	Additive
AA	$X = (0, 1)^T$	$X = 1$	$X = 1$	$X = 2$
Aa	$X = (1, 0)^T$	$X = 1$	$X = 0$	$X = 1$
aa	$X = (0, 0)^T$	$X = 0$	$X = 0$	$X = 0$

Assume a logistic regression model:

$$\log \left[\frac{\Pr(Y = 1|X)}{\Pr(Y = 0|X)} \right] = \beta_0 + \alpha C + \beta_1 X$$

where β_0 is the intercept, α is the coefficient for covariates C , and β_1 is the genetic effect-size (i.e., log(Odds-Ratio)).

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

Test Statistic

- Wald Test: $Z = \frac{\widehat{\beta}_1}{\text{Standard_Error}(\widehat{\beta}_1)} \sim N(0, 1)$ under H_0
- Chi-square Test: $X^2 = \frac{\widehat{\beta}_1^2}{\text{Var}(\widehat{\beta}_1)} \sim \text{Chi_Square}$ with $\text{df}=1$ under H_0
- How to obtain p-value?

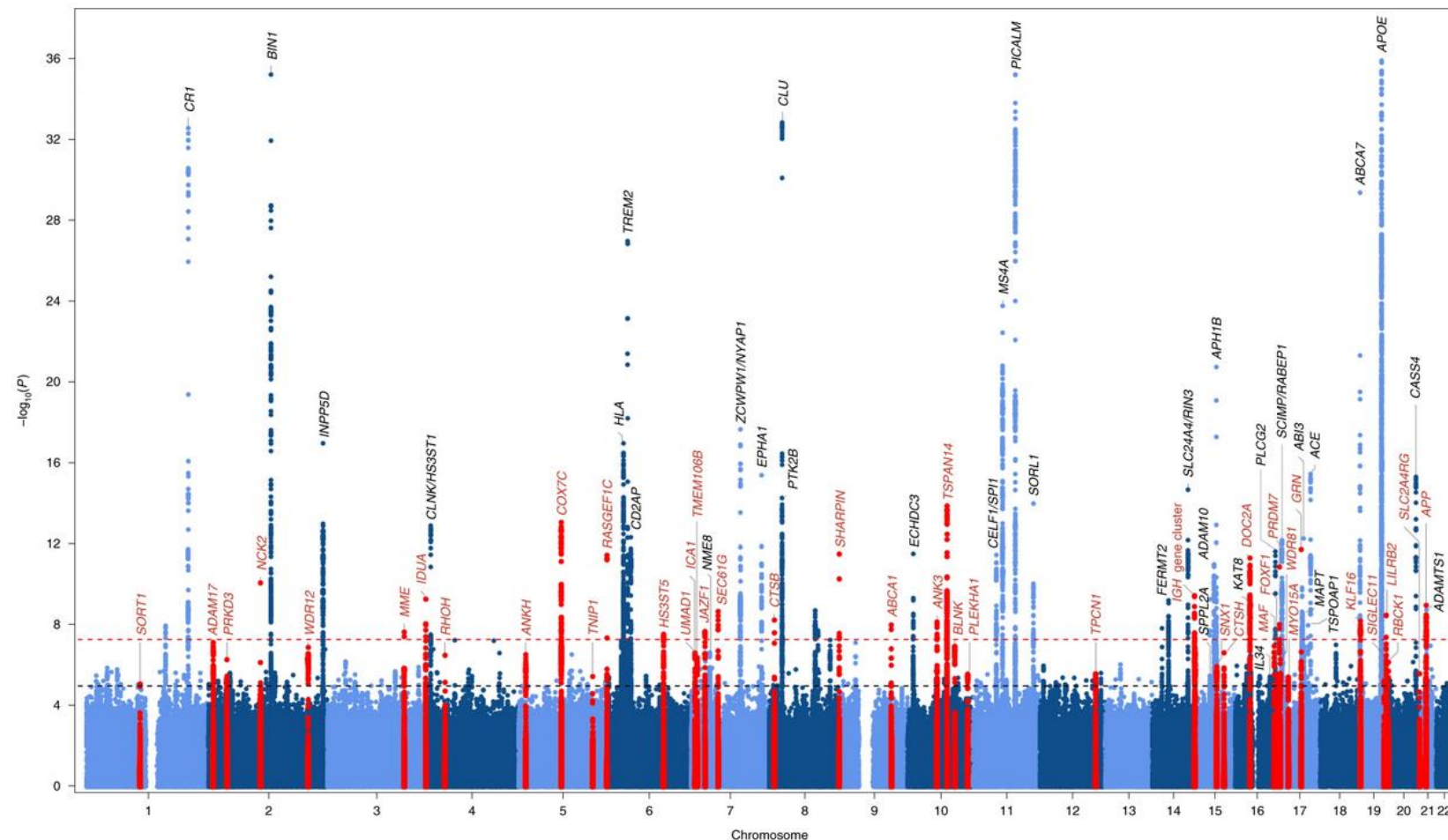
Visualize GWAS Results by Manhattan Plot

- Scatter plot of $-\log_{10}(\text{p-values})$ across all genome-wide variants
- Visualize signal peaks



Genome-wide Association Study (GWAS)

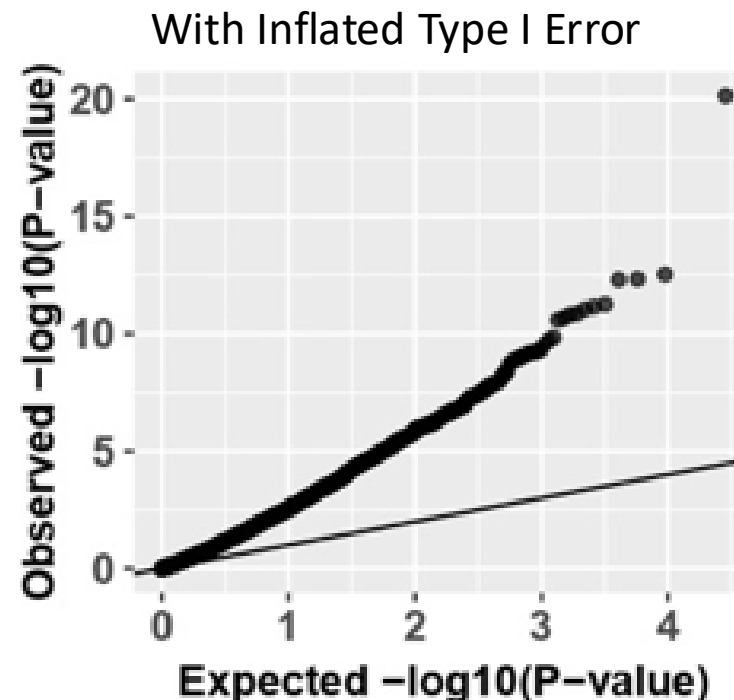
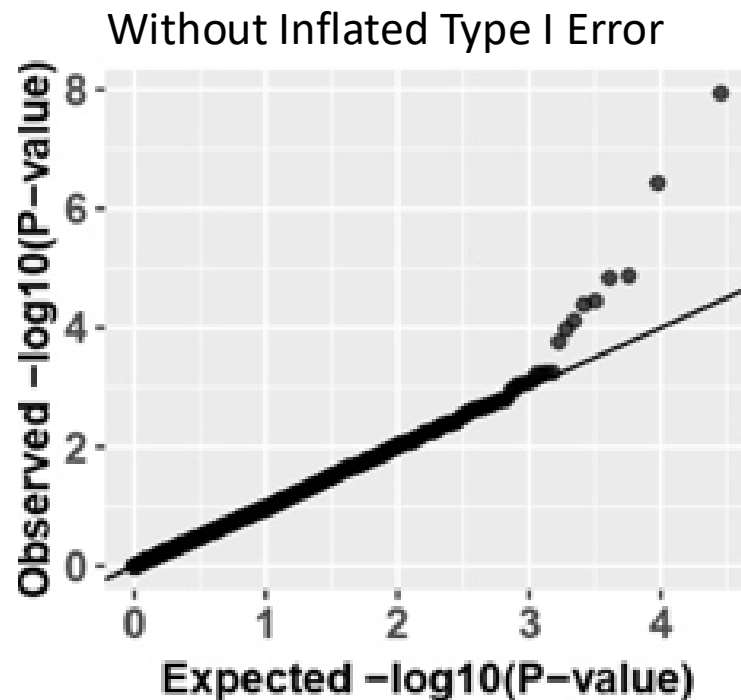
- Test the association between SNPs and the phenotype of interest
 - Independent single variant test for each SNP
 - **Summary data:** Effect sizes, Effect size standard deviations, Z-scores, P-values



GWAS of AD
dementia by
Bellenguez C. et
al. Nat Genetic.
2022

Check GWAS Results by Quantile-Quantile (QQ) Plot

- Obtained $-\log_{10}(\text{p-values})$ from GWAS
- Sort all $-\log_{10}(\text{p-values})$ from most significant to least
- Pair these with the expected values of order statistics of a $\text{Uniform}(0, 1)$ distribution
- Under NULL hypothesis (no association), p-values follow a $\text{Uniform}(0, 1)$ distribution



Genomic Control Factor

Genomic Control Factor is used to control for systematic inflation of type I error.

The idea is that the statistic T is inflated by an inflation factor λ (i.e., genomic control factor) so that

$$T \sim \lambda \chi_1^2$$

where λ can be estimated by

$$\hat{\lambda} = \text{median}(T_1, T_2, \dots, T_M)/0.456$$

- M is the number of independent tests, though in practice all tests are included.
- The denominator is the median of χ_1^2 distribution.
- $\hat{\lambda}$ should be 1 under H_0 .

Adjust GWAS results by Genomic Control Factor λ_{GC}

- Under null hypothesis (no association signal exists), p-values should follow a uniform distribution within (0, 1)
- Median p-value = 0.5 under null hypothesis, corresponding to chi-square statistic (df=1) value 0.456
- Find the actual median p-value from your GWAS, with corresponding chi-square statistic (df=1) value $\text{median}(T)$

$$\text{Genomic Control Factor: } \lambda_{GC} = \text{median}(T)/0.456$$

- Adjust your GWAS results by λ_{GC}
 - Scale your chi-square statistic test statistics (df=1) by λ_{GC}
 - Recalculate the corresponding GWAS p-values
 - Re-check QQ plot

GWAS Catalogue Results

2019 July

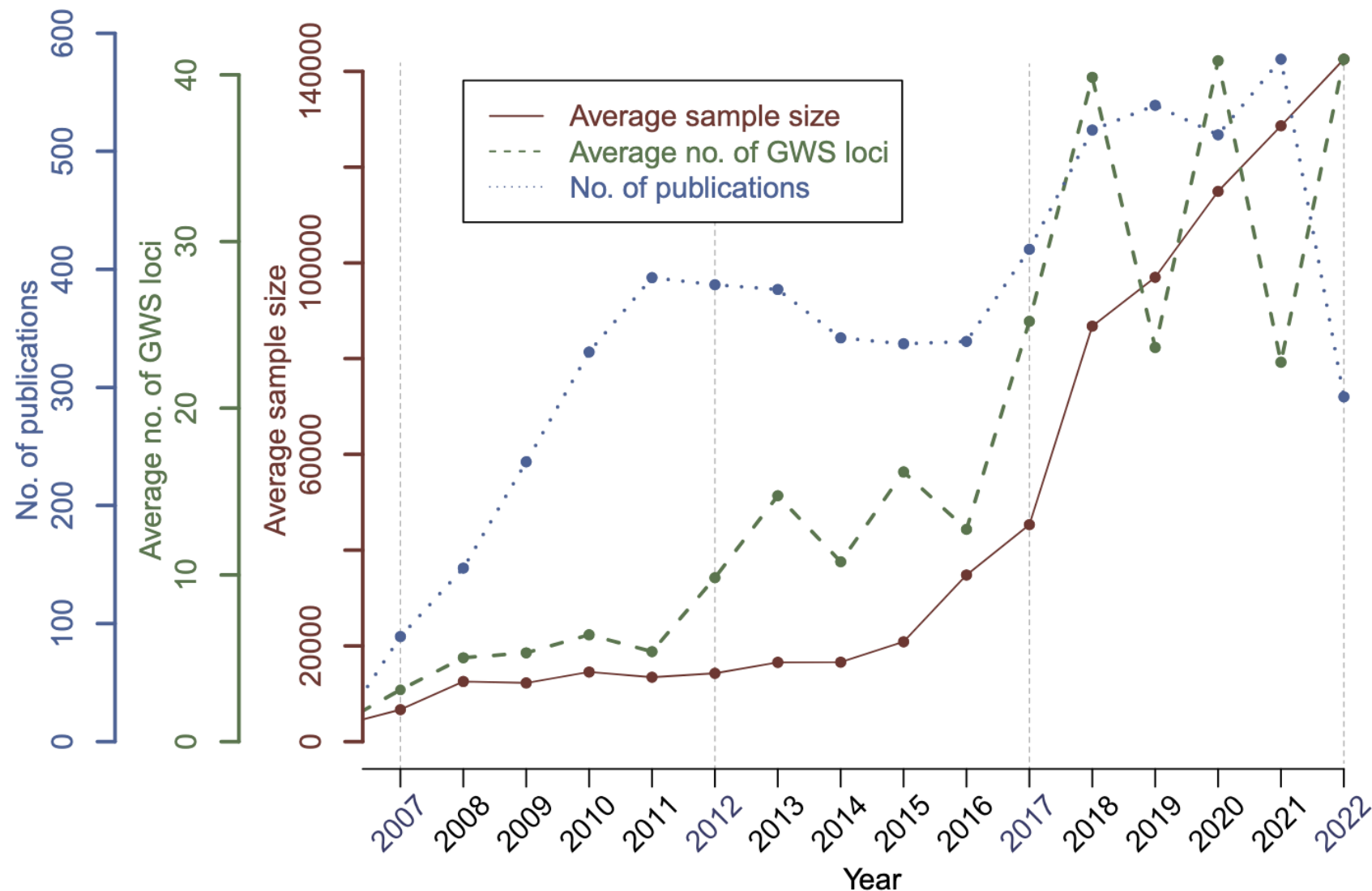
>157K Associations
from 4220 Publications



www.ebi.ac.uk/gwas

Challenge: Biological
Mechanism Underlying
GWAS Loci

- Mapping top significant SNP to the nearest gene?
- How to aggregate SNPs for Gene-based association tests?

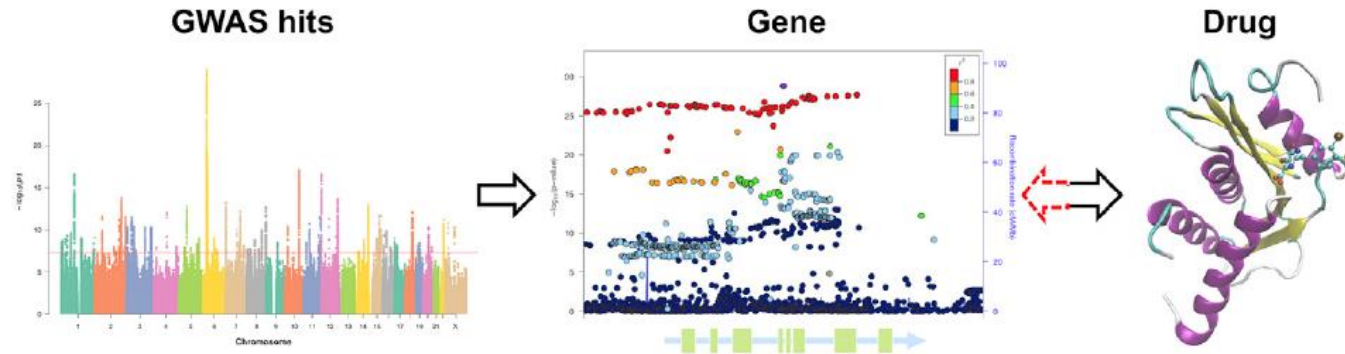


Abdellaoui A, et. al.
15 years of GWAS
discovery: Realizing
the promise. AJHG.
2023.

Figure 1. Average sample size and average number of genome-wide significant (GWS) loci per publication for each year during the 15 years history of GWAS discoveries

The data were extracted from 5,771 GWAS publications that used a genome-wide genotyping array and shared their summary statistics on GWAS Catalog before November 8, 2022.

Example links between GWAS discoveries and drug developments

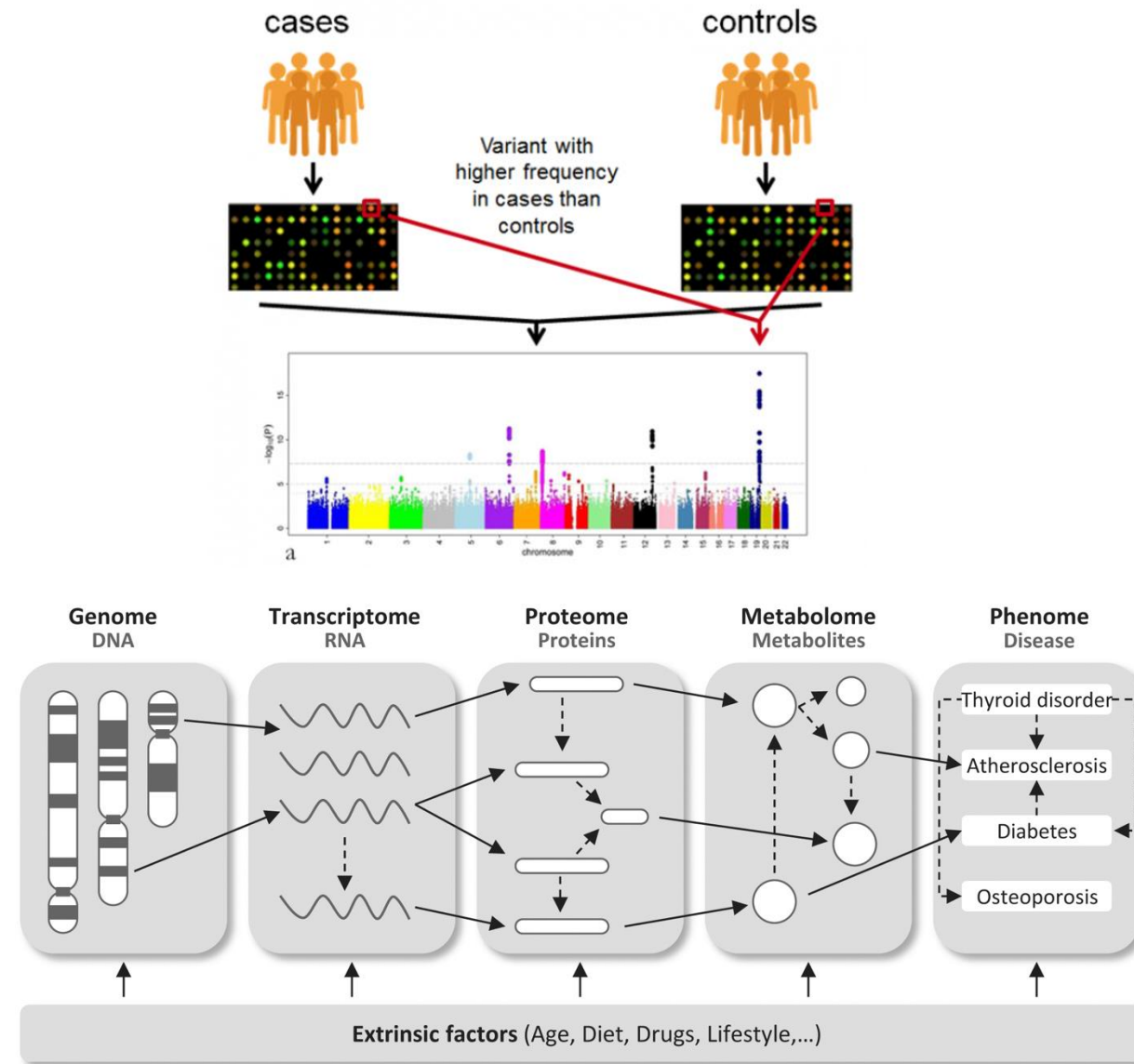


Trait	Gene with GWAS hits	Known or candidate drug
Type 2 Diabetes	<i>SLC30A8/KCNJ11</i>	ZnT-8 antagonists/Glyburide
Rheumatoid Arthritis	<i>PADI4/IL6R</i>	BB-CI-amidine/Tocilizumab
Ankylosing Spondylitis(AS)	<i>TNFR1/PTGER4/TYK2</i>	TNF-inhibitors/NSAIDs/fostamatinib
Psoriasis(Ps)	<i>IL23A</i>	Risankizumab
Osteoporosis	<i>RANKL/ESR1</i>	Denosumab/Raloxifene and HRT
Schizophrenia	<i>DRD2</i>	Anti-psychotics
LDL cholesterol	<i>HMGCR</i>	Pravastatin
AS, Ps, Psoriatic Arthritis	<i>IL12B</i>	Ustekinumab

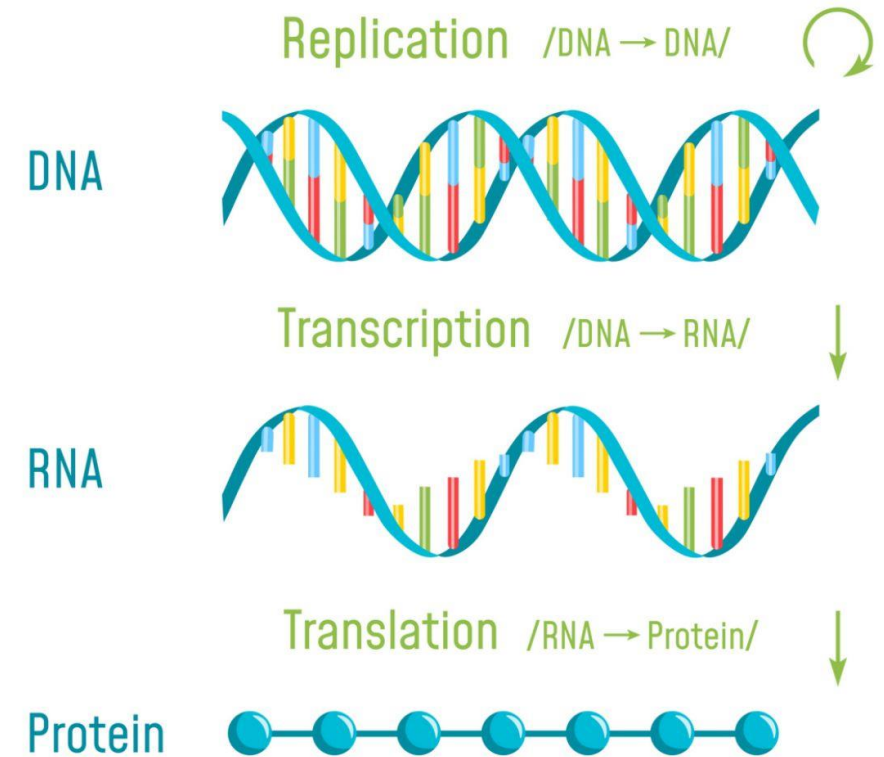
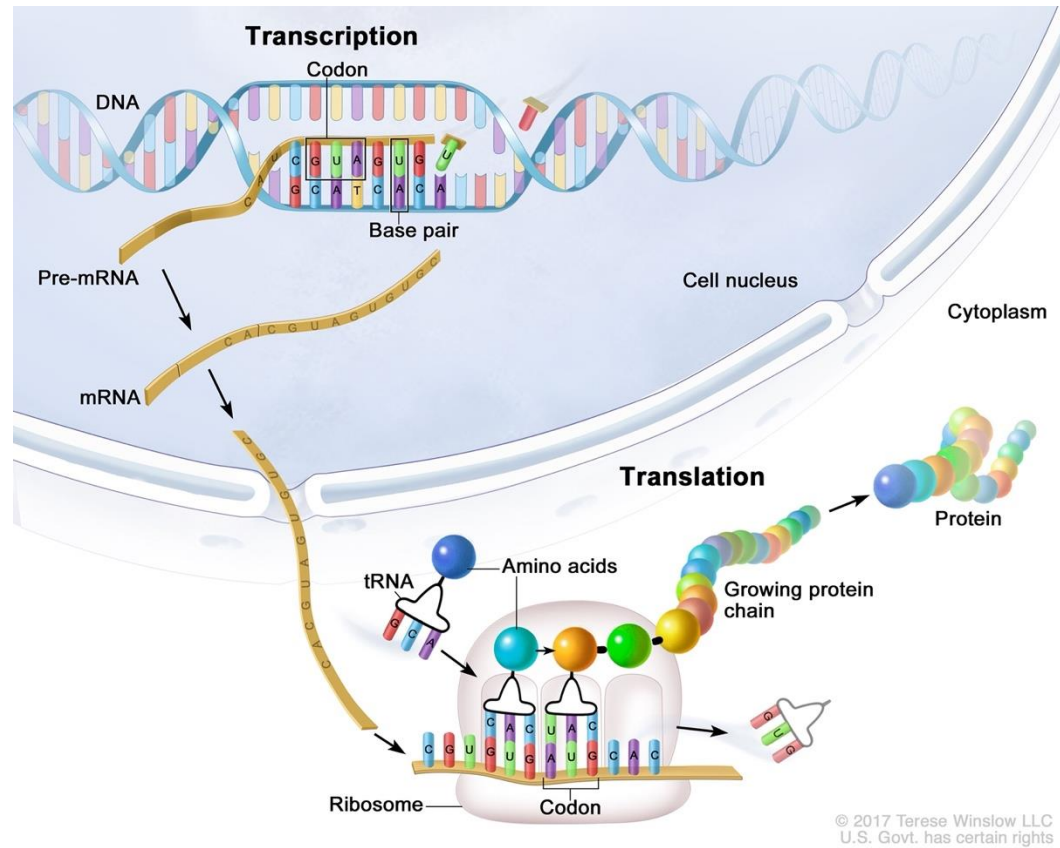
Visscher P.M.
et al. AJHG
2017.

Limitations by GWAS

- Many SNP 'hits' are in **non-coding regions** with unclear molecular mechanism
- Such SNPs often enriched for **expression quantitative trait loci (eQTLs)**
- Suggests **Genetically Regulated gene eXpression (GReX)** plays possible role in etiology of complex traits

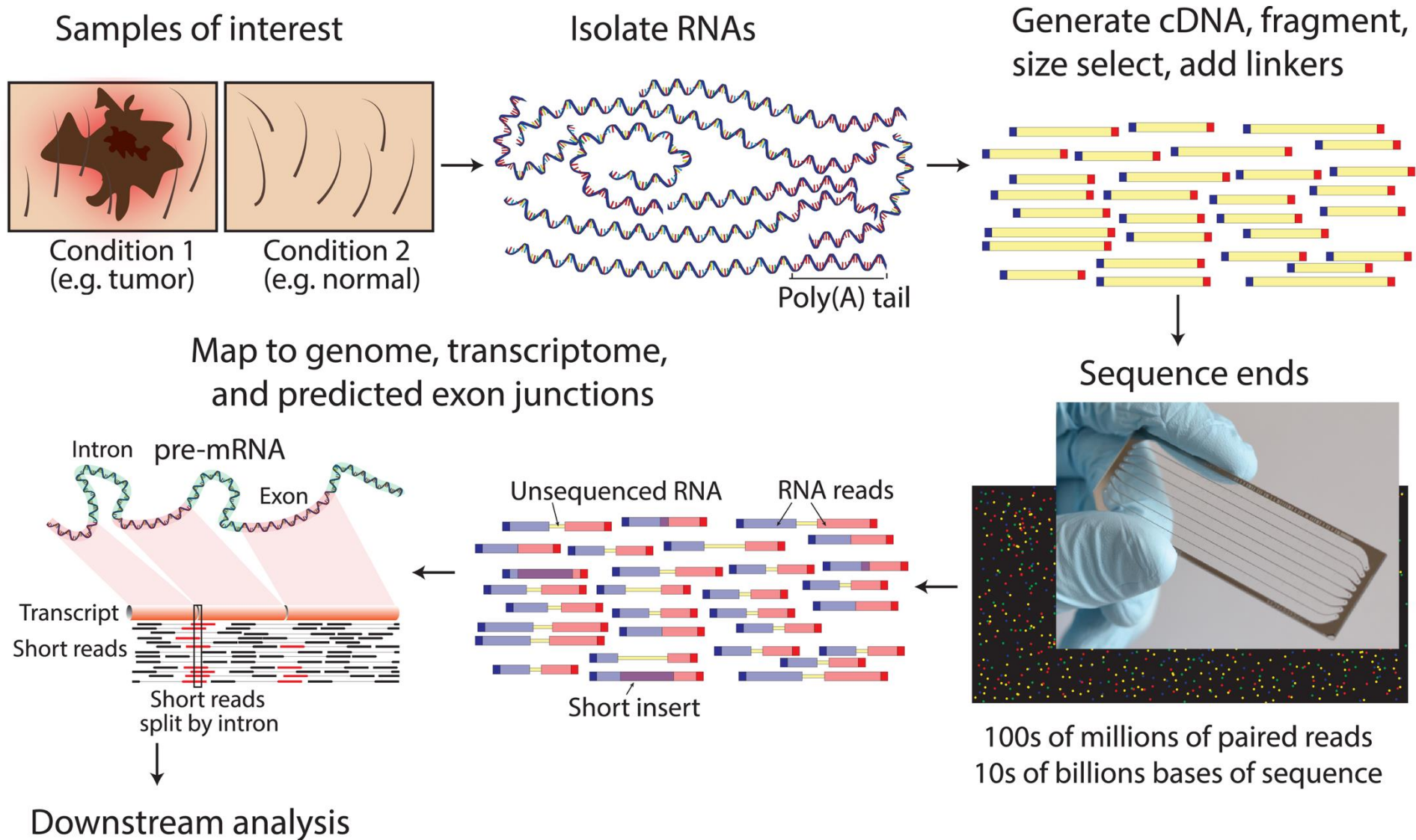


Transcription



<https://www.thoughtco.com/dna-transcription-373398>

Profile Gene Expression Levels by RNA-sequencing



Transcriptomic Data

- Gene expression Quantitative Traits
 - Profiled by RNA sequencing (RNA-seq)
 - TPM (Transcripts Per Kilobase Million) per gene
 - Divide the read counts by the length of each gene in kilobases. This gives you reads per kilobase (RPK).
 - Count up all the RPK values in a sample and divide this number by 1,000,000. This is your “per million” scaling factor.
 - Divide the RPK values by the “per million” scaling factor. This gives you TPM.
- 20K ~ 25K genes in human genome
- Tissue specific
 - Subject to relevant tissue availability
- Could be time specific
 - Vary over development or disease course
- Could be cell-type specific
 - Bulk RNA-seq vs. Single Cell RNA-seq

eQTL

- Consider the profiled gene expression levels as the quantitative trait E_g in the following single variant linear regression model:

$$E_g = \beta_0 + \alpha C + \beta_1 X + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

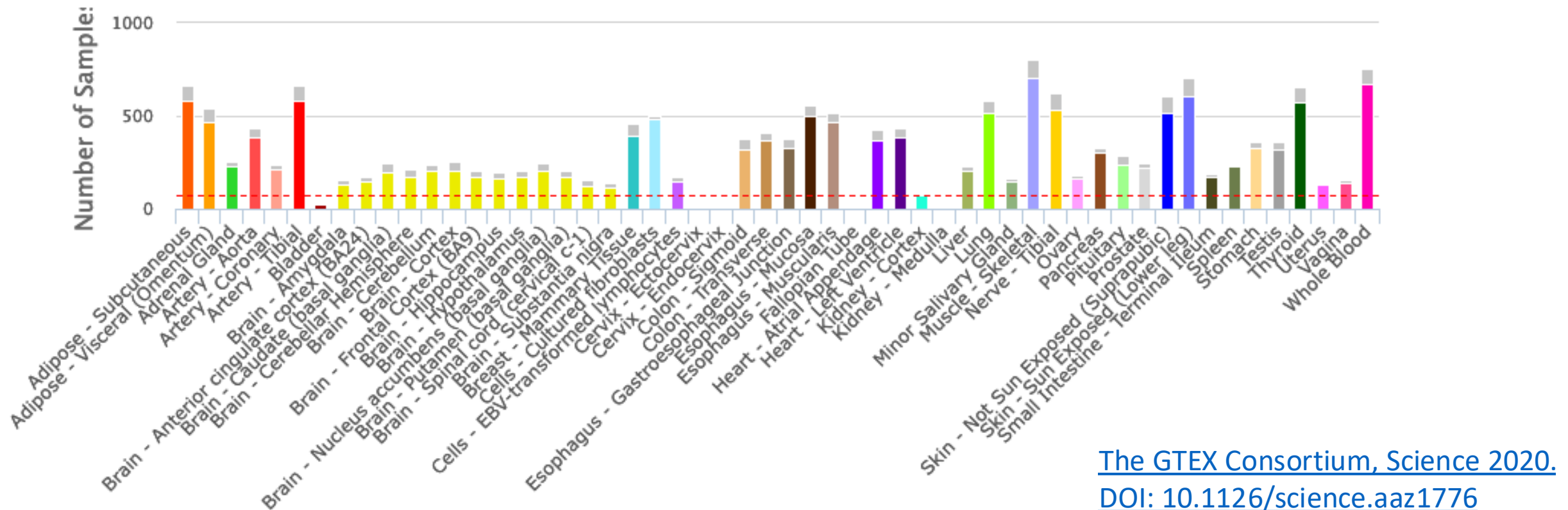
- X represents the genotype data (0, 1, 2) or dosage [0, 2] of the test SNP
 - C represents the confounding covariates or other environmental variables
 - ϵ represents the error term, other unknown factors
- eQTL : SNPs **significantly associated** with a gene expression quantitative trait ($H_0: \beta_1 = 0$ is significantly rejected) are referred as **expression Quantitative Trait Loci (eQTL)**
- Cis-eQTL : eQTL nearby the test gene (e.g., located within the +-1MB region of the transcription starting site; thousands cis-SNPs per gene).
- Trans-eQTL : eQTL distant from the test gene (e.g., located out of the +-1MB region of the transcription starting site, or on different chromosome; ~10M trans-SNPs per gene).

Genotype-Tissue Expression (GTEx) project

<https://www.gtexportal.org/home/>

V8 Release	# Tissues	# Donors	# Samples
Total	54	948	17382
With Genotype	54	838	15253
Has eQTL Analysis*	49	838	15201

* Number of samples with genotype ≥ 70



[The GTEx Consortium, Science 2020.](#)
[DOI: 10.1126/science.aaz1776](#)

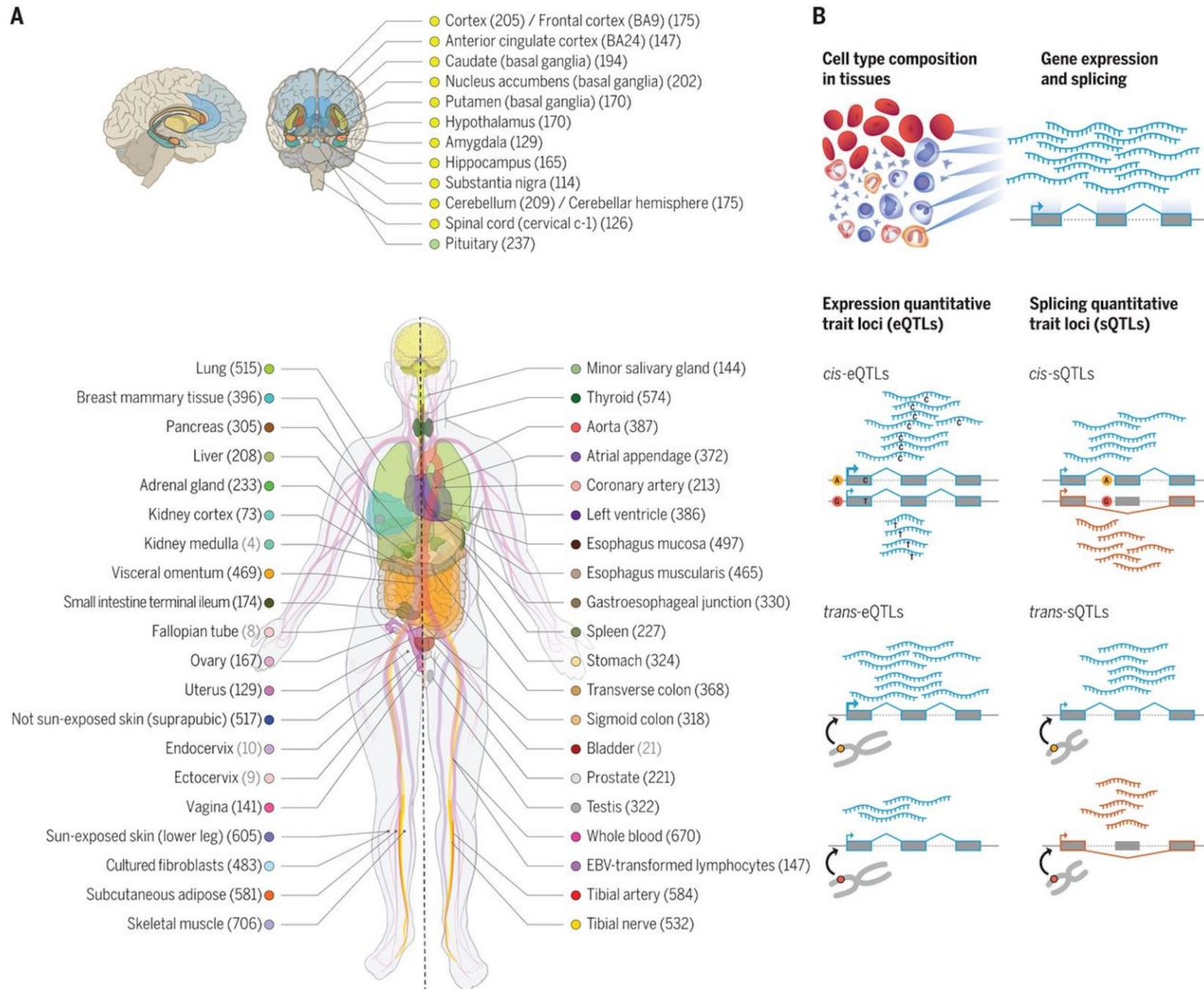


Fig. 1 Sample and data types in the GTEx v8 study.

(A) Illustration of the 54 tissue types examined (including 11 distinct brain regions and two cell lines), with sample numbers from genotyped donors in parentheses and color coding indicated in the adjacent circles. Tissues with 70 or more samples were included in QTL analyses.

(B) Illustration of the core data types used throughout the study. Gene expression and splicing were quantified from bulk RNA-seq of heterogeneous tissue samples, and local and distal genetic effects (*cis*-QTLs and *trans*-QTLs, respectively) were quantified across individuals for each tissue.

[The GTEx Consortium, Science 2020.
DOI: 10.1126/science.aaz1776](https://doi.org/10.1126/science.aaz1776)

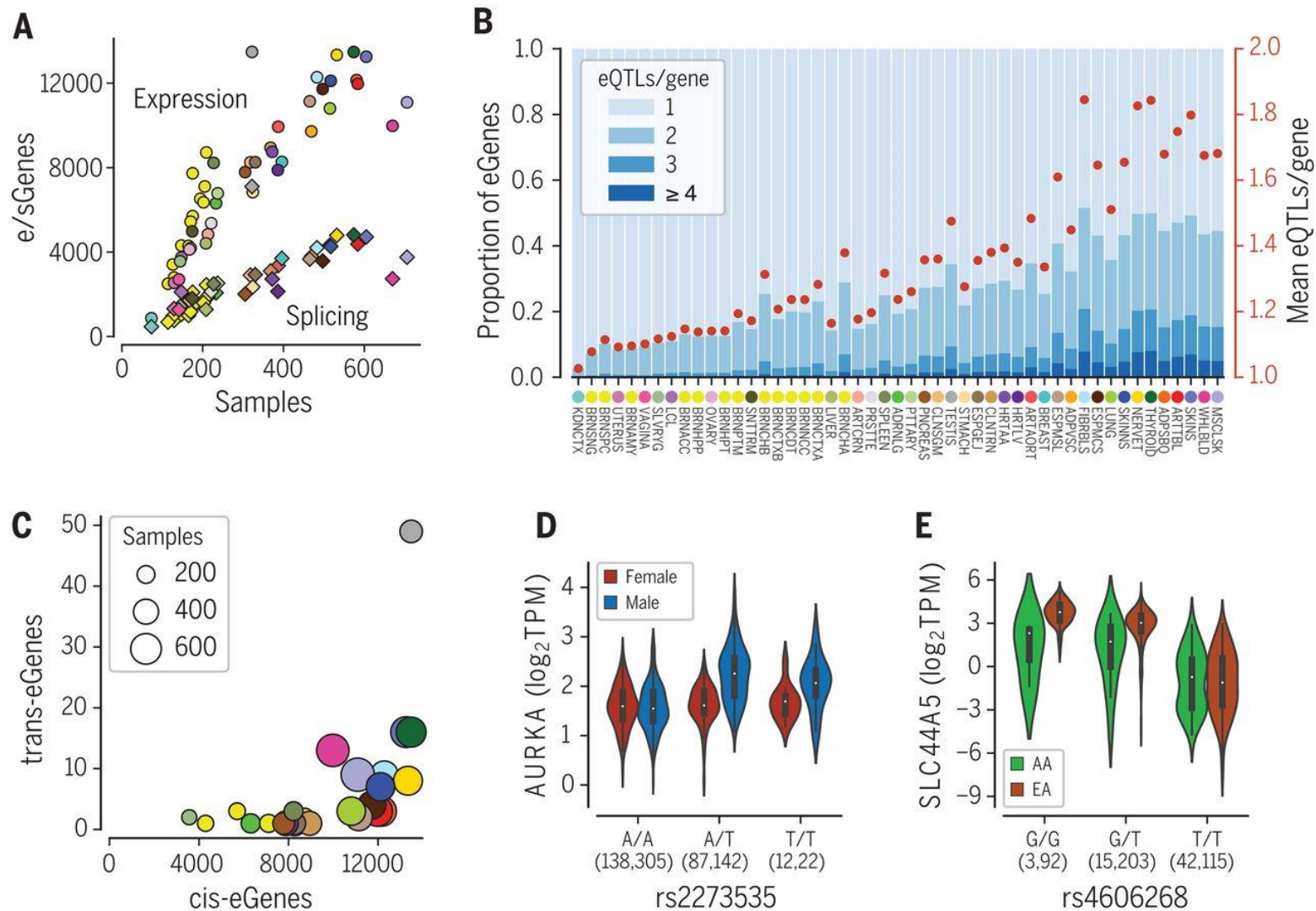


Fig 2. QTL discovery.

(A) The number of genes with a cis-eQTL (eGenes) or cis-sQTL (sGenes) per tissue, as a function of sample size. See [Fig. 1A](#) for the legend of tissue colors.

(B) **Allelic heterogeneity** of cis-eQTLs depicted as proportion of eGenes with one or more independent cis-eQTLs (blue stacked bars; left y axis) and as a mean number of cis-eQTLs per gene (red dots; right y axis). The tissues are ordered by sample size.

(C) The number of genes with a trans-eQTL as a function of the number of cis-eGenes.

(D) **Sex-biased** cis-eQTL for *AURKA* in skeletal muscle, where rs2273535-T is associated with increased *AURKA* expression in males ($P = 9.02 \times 10^{-27}$) but not in females ($P = 0.75$).

(E) **Population-biased** cis-eQTL for *SLC44A5* in esophagus mucosa (aFC = -2.85 and -4.82 and in African Americans (AA) and European Americans (EA), respectively; permutation P value = 1.2×10^{-3}).

TPM, transcripts per million.

[The GTEX Consortium, Science 2020. DOI: 10.1126/science.aaz1776](#)

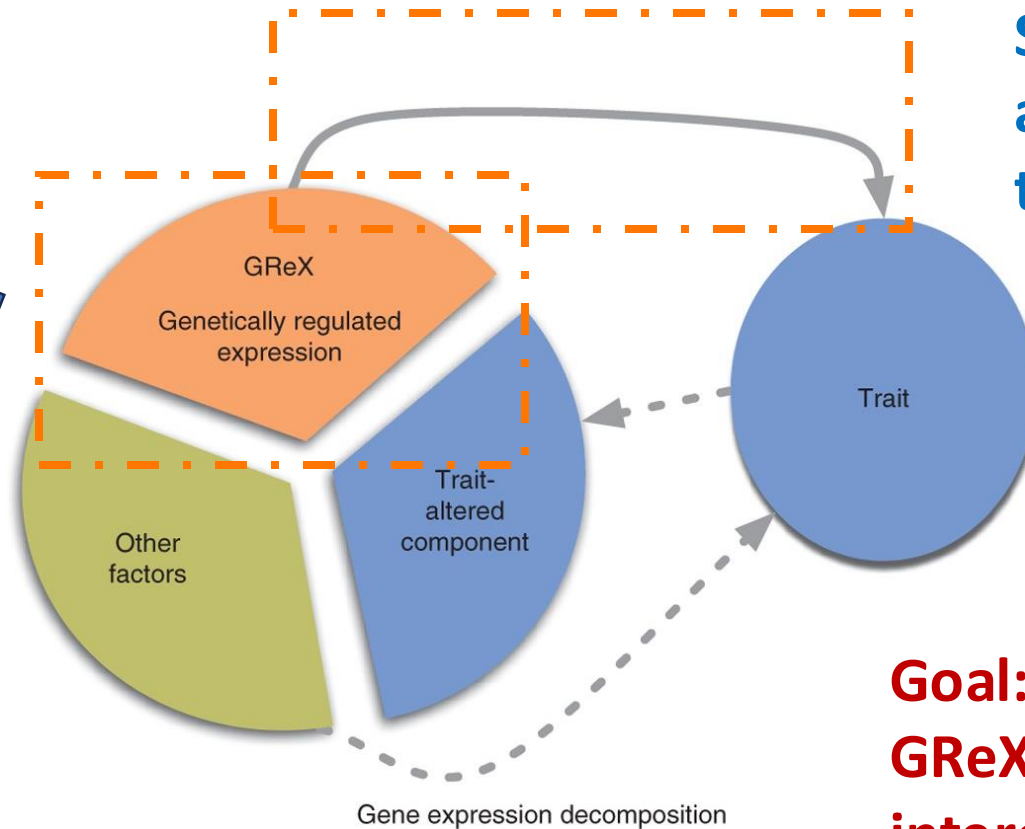
Allelic heterogeneity: More than one variant at the same locus independently affects gene expression.

Transcriptome-wide Association Study (TWAS)

Step 1: Identify the part of gene expression that are regulated by genetic variants (GReX)

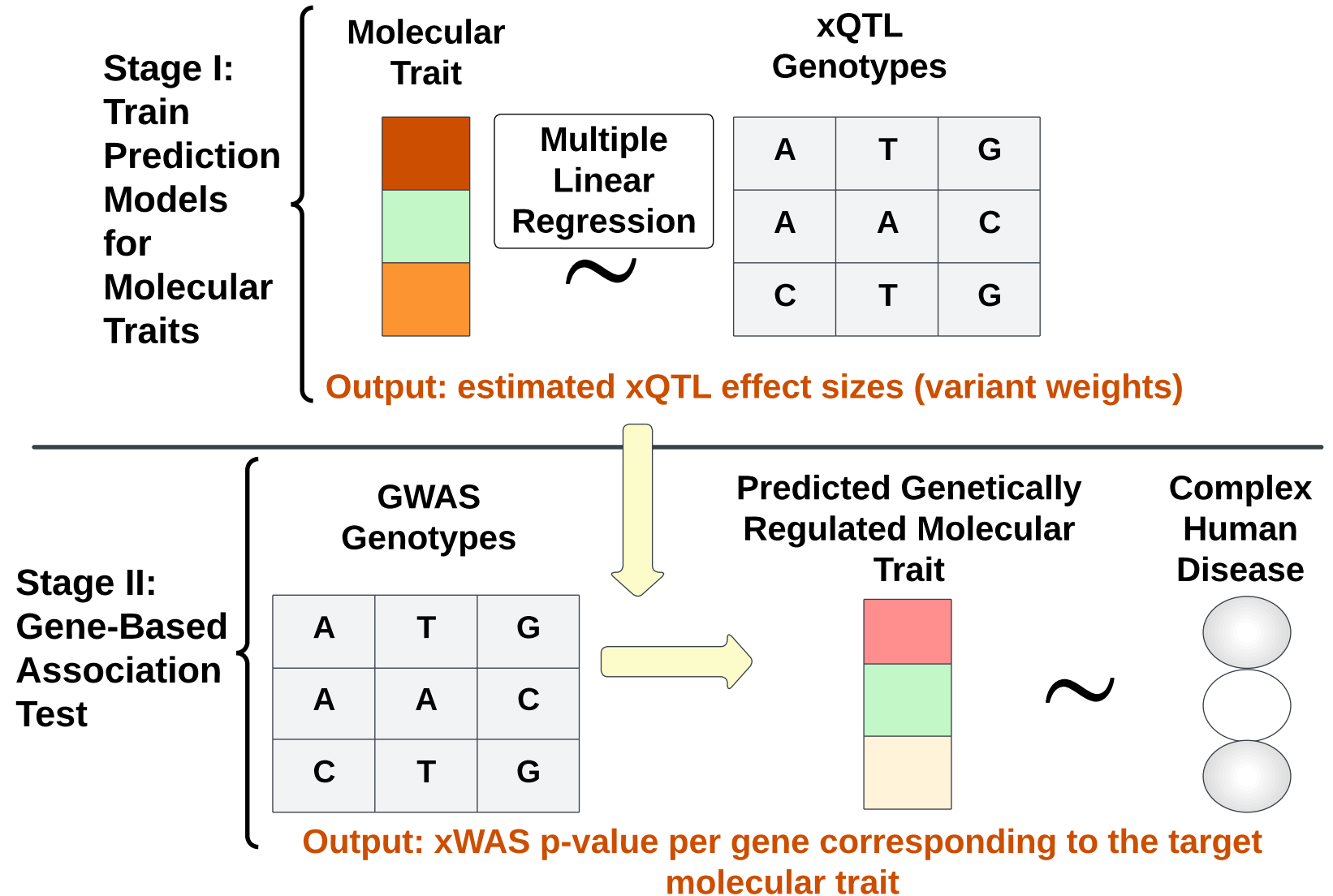
Step 2: Test the association between the trait and GReX

Goal: Identify risk genes with GReX associated with the trait of interest.



Standard Two-Stage TWAS Framework

- Integrates transcriptomic data (or other molecular trait) with GWAS data
- Leverage publicly available transcriptomic (or other omics) data
- Gene-based association test



Standard Two-Stage TWAS

- 1) Train GReX prediction models and estimate “eQTL” effect sizes with reference panel

$$\mathbf{E}_g = \mathbf{X}_{n \times p} \mathbf{w}_{p \times 1} + \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon} \sim N(0, \sigma_{\varepsilon}^2 \mathbf{I})$$

- $\mathbf{E}_g$, Gene Expression adjusted for confounding covariates, with mean 0
- $\mathbf{X}_{n \times p}$ Reference Genotypes of cis-SNPs (within +/-1Mb around the test gene region), with column means 0
- $\mathbf{w}_{p \times 1}$: “eQTL” effect sizes, i.e., “eQTL” weights

- 2) Test gene-based association with independent GWAS test data

$$\mathbf{Y}_{\text{test}} \sim \beta * \widehat{\mathbf{GReX}}$$

- $\widehat{\mathbf{GReX}} = \mathbf{X}_{\text{test}} \hat{\mathbf{w}}_{p \times 1}$: Predicted GReX in test cohort, with genotype data $\mathbf{X}_{\text{test}}$
- $\mathbf{Y}_{\text{test}}$: Phenotype of interest;
- Test $H_0: \beta = 0$

TWAS with GWAS Summary Data

- GWAS summary data:
 - GWAS Z-scores ($Z_l, l = 1, \dots, m$) for test SNPs with non-zero eQTL weights $\hat{w}$, derived from single variant tests
 - Genotype correlation and variance-covariance matrices ($\mathbf{V}$) derived from reference genotype data
- TWAS Z-score statistics are given by:

$$\begin{aligned}\tilde{Z}_{g,\text{FUSION}} &= \frac{\sum_{l=1}^m (\hat{w}_l Z_l)}{\sqrt{\hat{\mathbf{w}}' \mathbf{V} \hat{\mathbf{w}}}}, \quad \mathbf{V} = \text{Corr}(\mathbf{G}_0) \\ \tilde{Z}_{g,\text{S-PrediXcan}} &= \frac{\sum_{l=1}^m (\hat{w}_l \hat{\sigma}_l Z_l)}{\sqrt{\hat{\mathbf{w}}' \mathbf{V} \hat{\mathbf{w}}}}, \quad \hat{\sigma}_l^2 = \text{Var}(\mathbf{G}_{0,1}), \quad \mathbf{V} = \text{Cov}(\mathbf{G}_0).\end{aligned}$$

Both TWAS Z-score Statistics are Equivalent with Centered and Standardized Gene Expression and Genotypes in the Reference Transcriptomic Panel

TWAS

- Essentially “functional” gene-based association test
 - Aggregate associations of multiple test SNPs per target gene, with SNP weights (eQTL effect sizes) derived from reference transcriptomic data
 - Improve power than single variant tests
 - Improve power than traditional gene-based association test with equal weights for test SNPs
- Better informing the underlying biological mechanism of identified risk genes than GWAS

TWAS Tools Estimate eQTL Weights Assuming Different Statistical Model

- **PrediXcan** (Gamazon ER et al., Nat Genetics, 2015): Use penalized linear regression model with Elastic-Net penalty
- **FUSION** (Gusev et al., Nat. Genetics, 2016): Select best model out of PrediXcan/Elastic-Net, LASSO, best unbiased linear predictor (BLUP), single variant model with Top eQTL (Top1), and Bayesian sparse linear mixed model (BSLMM) with the highest cross-validation (CV) R^2
- **TIGAR** (Nagpal S. et al., AJHG, 2019): Use nonparametric Bayesian Dirichlet Process Regression (DPR) model

PrediXcan (Gamazon ER et. al., Nat Genetics, 2015)

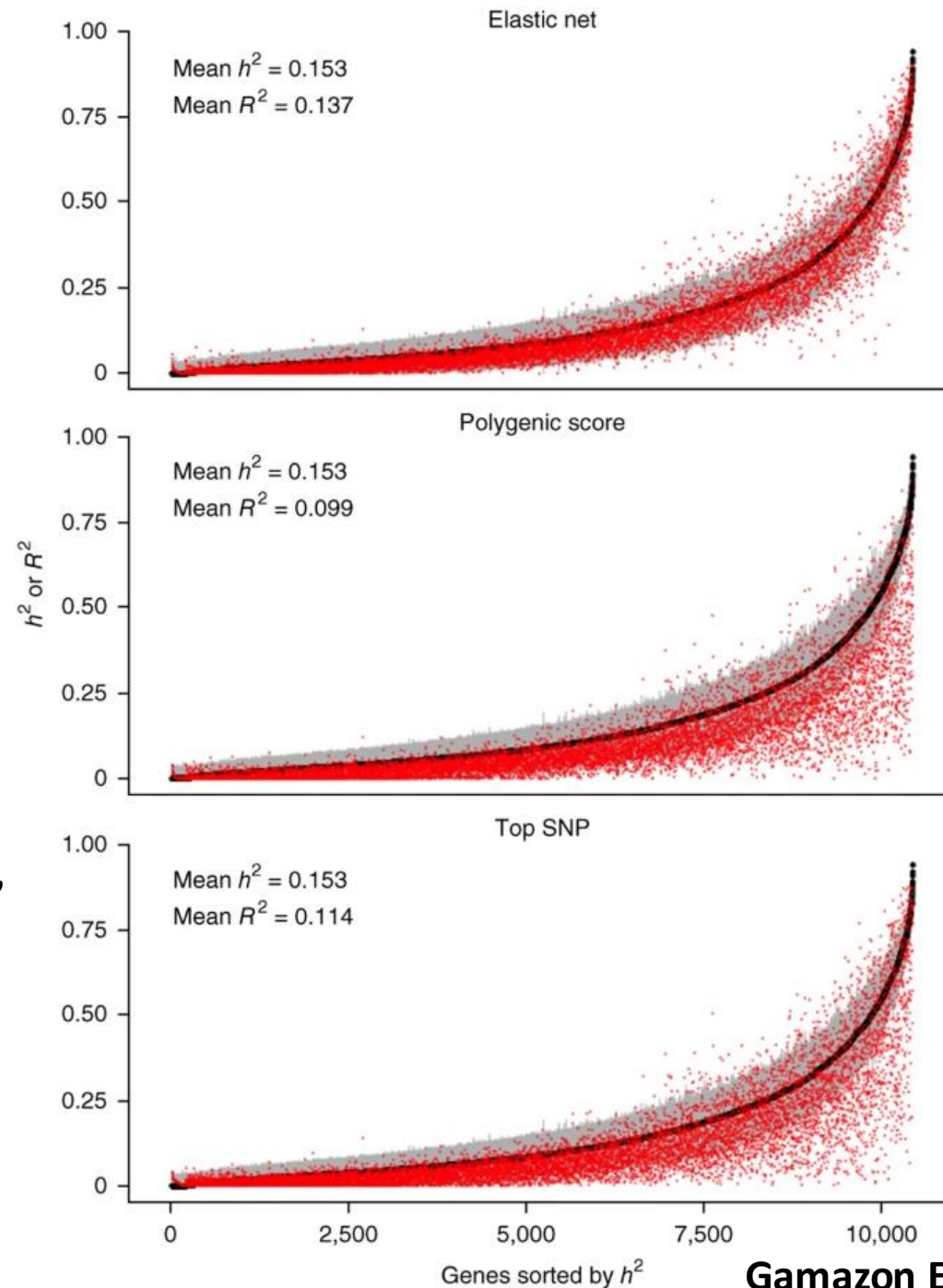
- Use Elastic-Net penalty with equal proportions of Ridge (L2) and LASSO (L1) penalties
- Minimize L2 norm of the difference between gene expression E_g and the genetically regulated component $X\mathbf{w}$

$$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmin}} \left(\|\mathbf{E}_g - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \left(\alpha \|\mathbf{w}\|_1 + \frac{1}{2} (1 - \alpha) \|\mathbf{w}\|_2^2 \right) \right)$$

- $\|\cdot\|_2$ denotes L_2 norm, $\|\cdot\|_1$ denotes L_1 norm, $\alpha \in [0, 1]$ denotes the proportion of L_1 penalty, and λ denotes the penalty parameter
- Take $\alpha = 0.5$
- Tune the penalty parameter λ by a 5-fold cross validation
- Often estimates a sparse or null eQTL model

10-fold CV R^2 versus expression heritability estimates

Compare Three
Models (Elastic net,
Polygenic score,
Top eQTL)



- Red dots: Prediction R^2
- Black dots: expression heritability estimates
- Gray lines: 95% confidence interval for expression heritability estimates

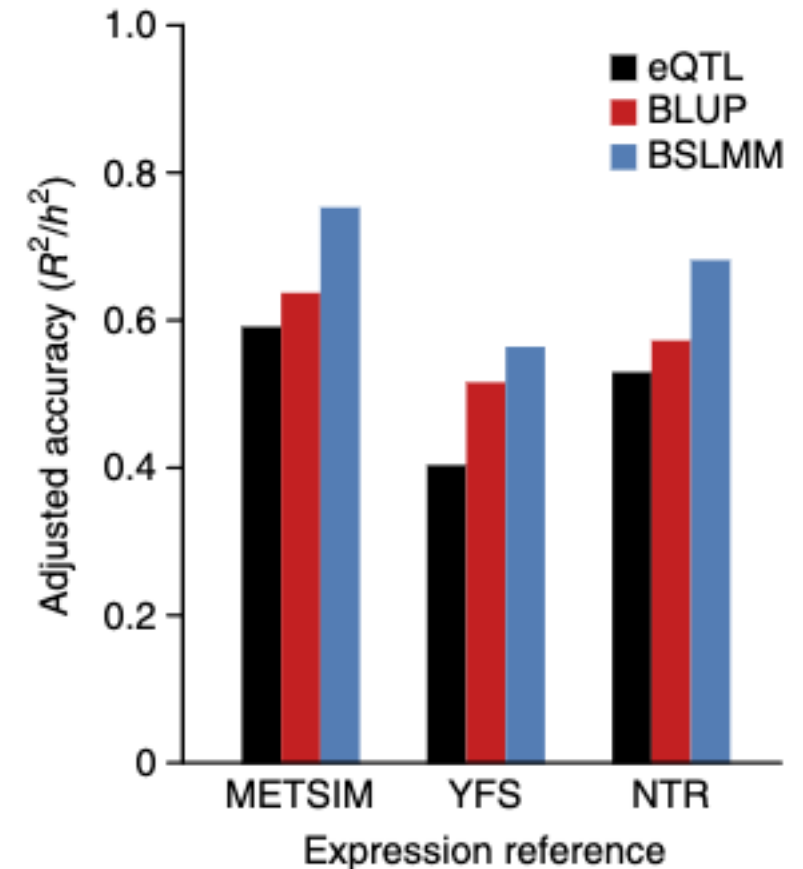
FUSION (Gusev et al., Nat. Genetics, 2016)

- Compared Top1 (most significant cis-eQTL), BLUE (standard least square estimate), and BSLMM (Zhou et al. PLOS Genetics, 2013).
- BSLMM (Zhou et al. PLOS Genetics, 2013)

$$\begin{aligned} E_g &= Xw + \epsilon, & \epsilon &\sim N(0, I\tau^{-1}); \\ w_i &\sim \pi N\left(0, \frac{\sigma_a^2 + \sigma_b^2}{\rho\tau}\right) + (1 - \pi)N\left(0, \frac{\sigma_b^2}{\rho\tau}\right) \end{aligned}$$

$\pi=0$: Equivalent to Linear Mixed Model, BLUE

$\sigma_b^2=0$: Equivalent to Bayesian variable selection regression model (BVSR, sparse model), assuming the spike-and-slab (i.e., point-normal) prior



Gusev et al., Nat. Genetics, 2016

TIGAR: Transcriptome-Integrated Genetic Association Resource (Nagpal et. al., AJHG 2019)

- Use nonparametric Bayesian Dirichlet Process Regression (DPR) model

$$\mathbf{E}_g = \mathbf{X}_{n \times p} \mathbf{w}_{p \times 1} + \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon} \sim N(0, \sigma_{\varepsilon}^2 \mathbf{I}), \sigma_{\varepsilon}^2 \sim IG(a_{\varepsilon}, b_{\varepsilon})$$

$$w_i \sim N(0, \sigma_{\varepsilon}^2 \sigma_w^2), \sigma_w^2 \sim D, D \sim DP(IG(a, b), \xi), i = 1, \dots, p$$

Base distribution

Concentration parameter

- Estimate eQTL effect sizes $w_{p \times 1}$ by Monte Carlo Markov Chain (MCMC) algorithm or Variational Bayesian Approximation.

Nonparametric Bayesian DPR Model

Another intuitive way of viewing this nonparametric model

- σ_w^2 can be viewed as a Latent variable
- Integrating out σ_w^2 will induce a Nonparametric prior distribution on w_i
- Equivalent to a normal mixture model for w_i

Infinite Gaussian Mixture Model

$$w_i \sim \pi_0 N(0, \sigma_\varepsilon^2 \sigma_0^2) + \sum_{k=1}^{+\infty} \pi_k N(0, \sigma_\varepsilon^2 (\sigma_k^2 + \sigma_0^2));$$

$$\pi_k = v_k \prod_{l=0}^{k-1} (1 - v_l), \quad v_k \sim \text{Beta}(1, \xi), \quad \xi \sim \text{Gamma}(a_\xi, b_\xi);$$

$$\sigma_k^2 \sim \text{IG}(a_k, b_k), \quad k = 0, 1, \dots, +\infty.$$

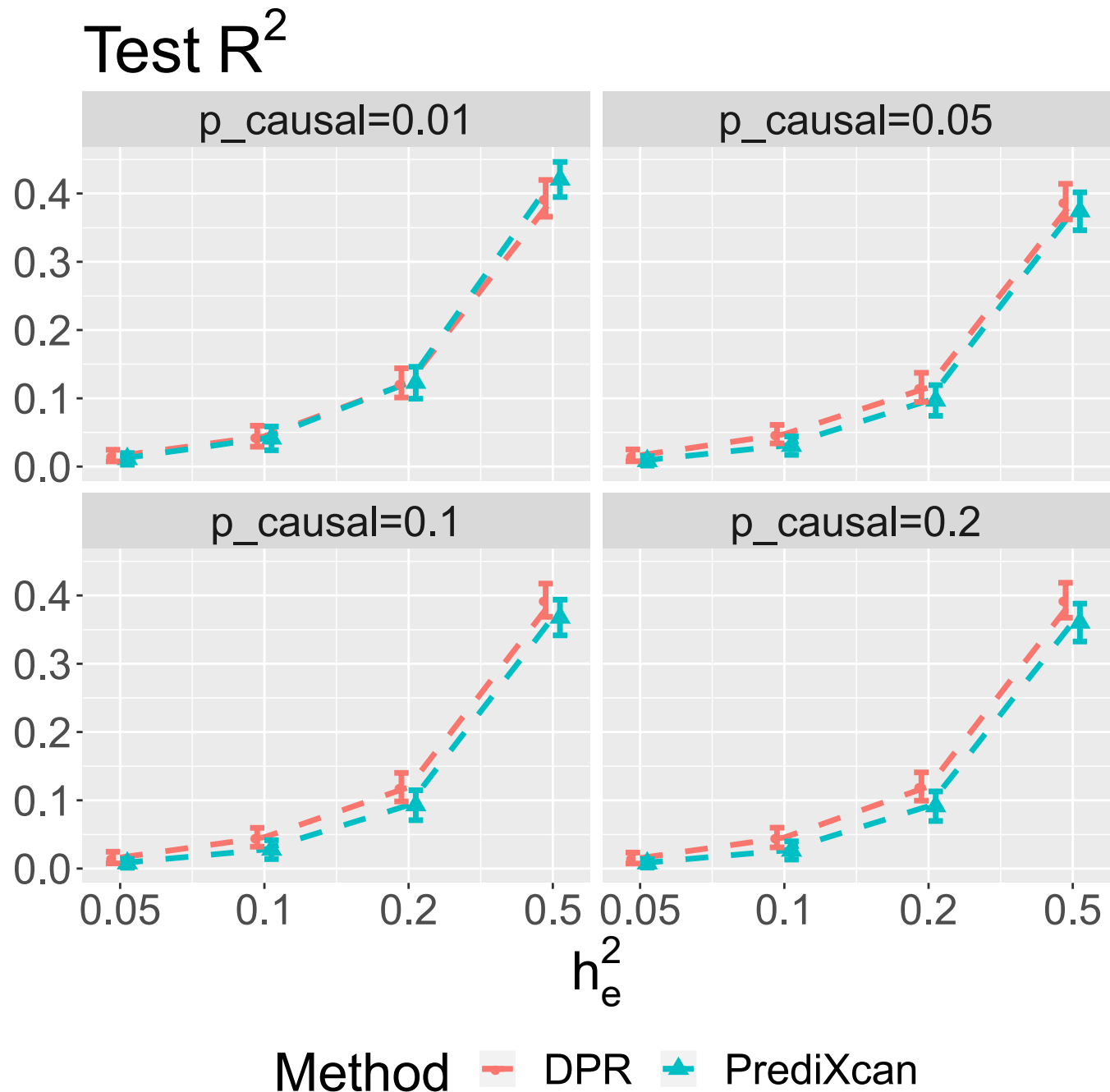
Advantages by using the Nonparametric Bayesian DPR model

- Assume infinitesimal eQTL model: all analyzed cis-SNPs contribute to the gene expression
- Include parametric Bayesian variable selection regression model (BVSR), and Bayesian sparse linear mixed model (BSLMM) as special cases
- Estimates gene expression prediction models for more genes, with higher average CV R^2 , than PrediXcan/Elastic-Net method
- Improves TWAS power

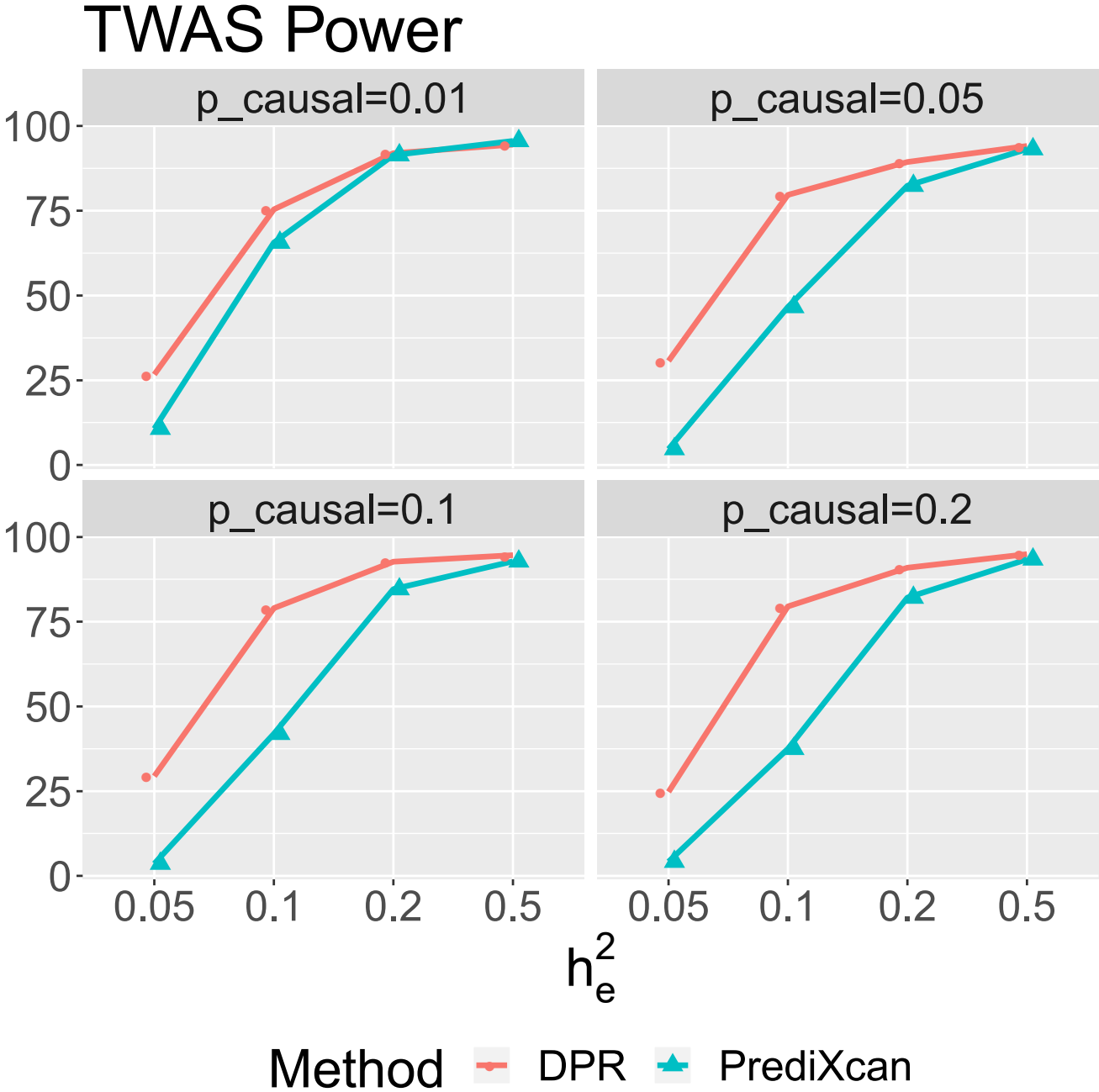
Simulation Study Design

- Use the real genotype data of gene *ABCA7* with 2799 cis-SNPs with $MAF > 5\%$ and $HWP > 10^{-5}$
- Training sample size (100, 300, 499), test sample size 1200
- Consider scenarios with various proportion of causal SNPs for gene expression, $p_{causal} = (0.01, 0.05, 0.1, 0.2)$
- Consider scenarios with various gene expression heritability and phenotype heritability, $(p_e^2, p_h^2) = ((0.05, 0.8), (0.1, 0.5), (0.2, 0.25), (0.5, 0.1))$
- Compare PrediXcan and DPR methods with respect to gene expression prediction R^2 and TWAS power

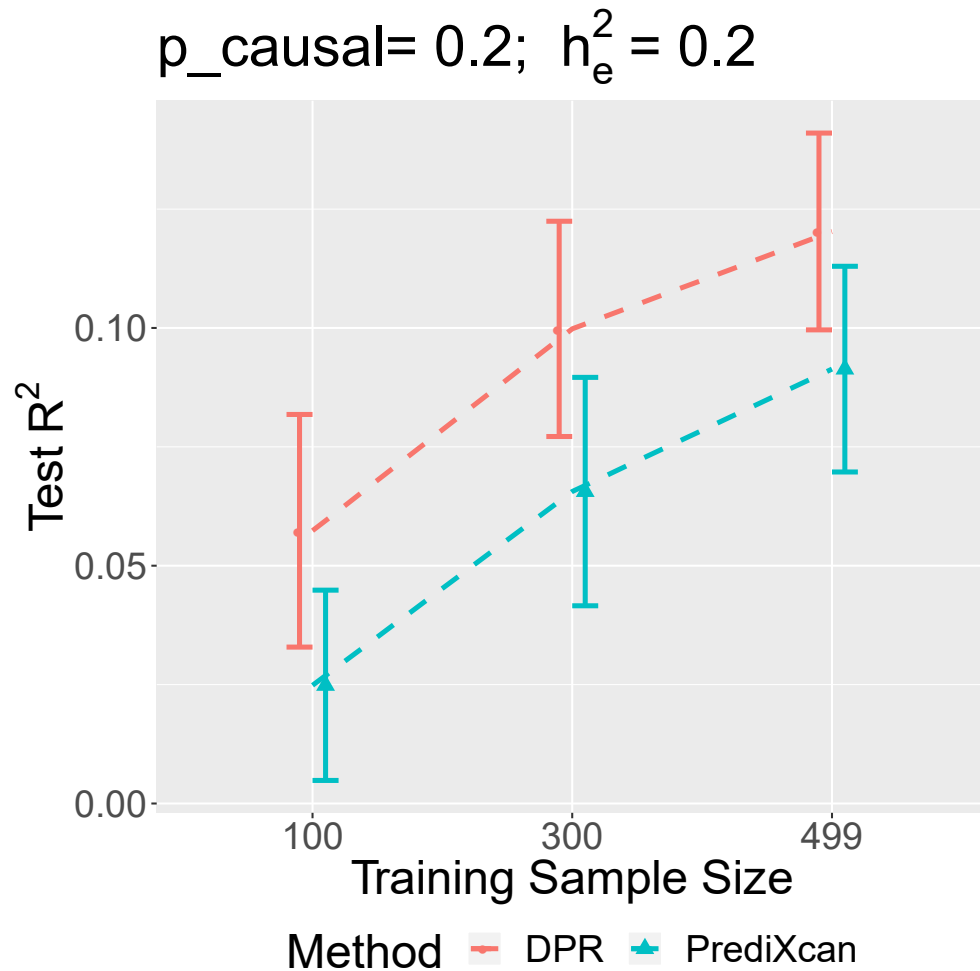
Gene Expression Prediction R^2 on Test Data



TWAS Power Comparison with Test Data



TWAS Power Increases as Training Sample Size Increases

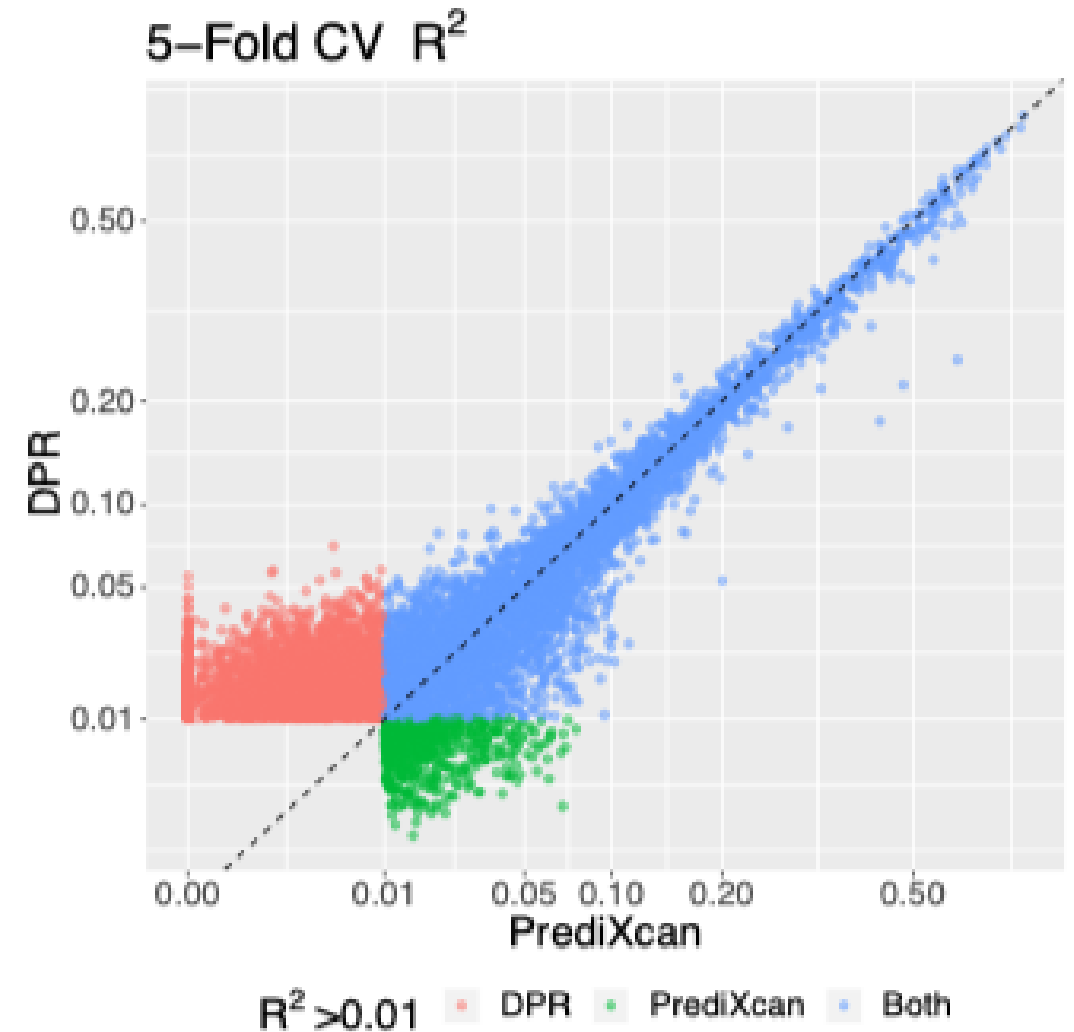


TWAS of Alzheimer's Disease Dementia

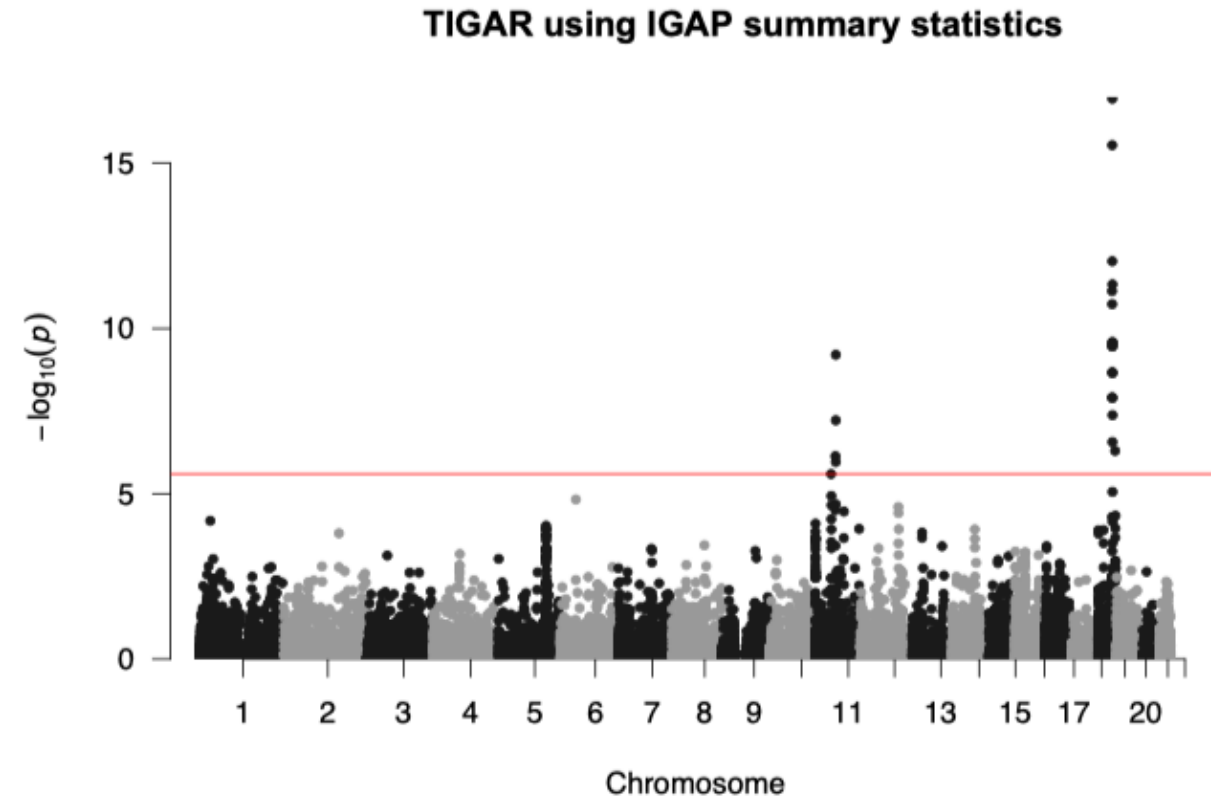
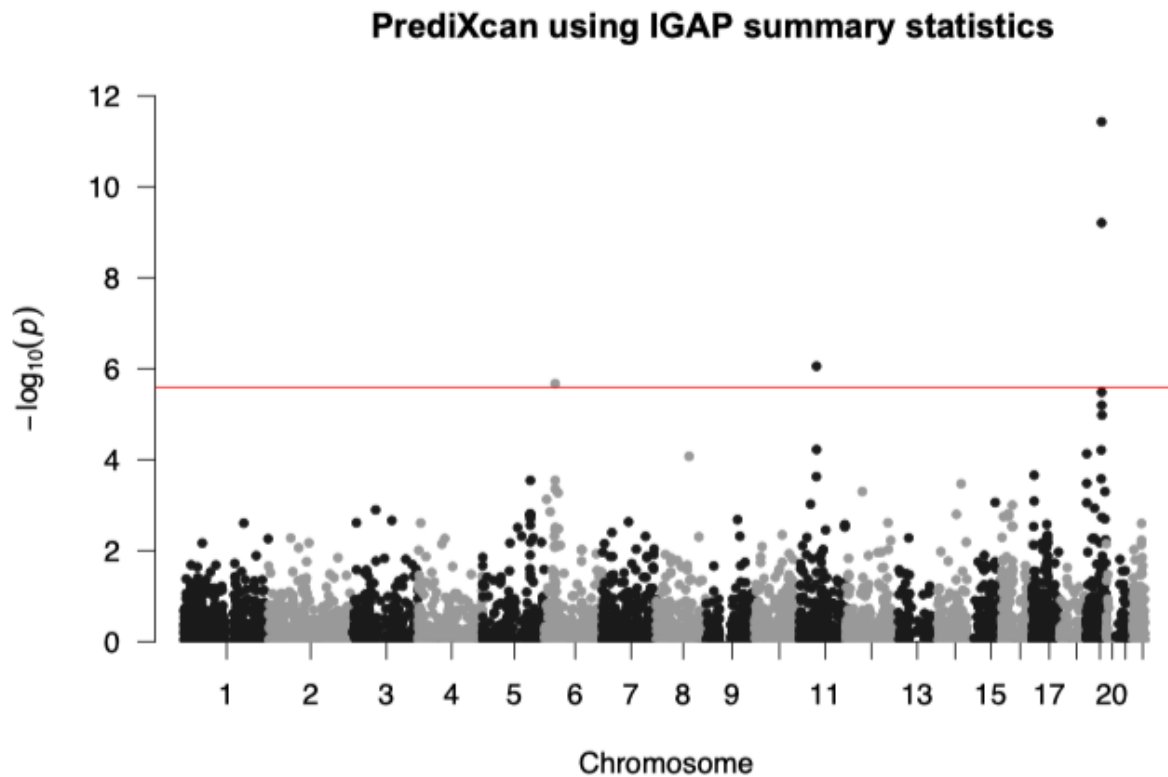
- Reference Data
 - Transcriptomic data (14K genes, RNAseq): 499 dorsolateral prefrontal cortex tissue from post-mortem brains from ROS/MAP cohorts
 - Genotype data for the same samples with profiled transcriptomic data
 - European ancestry
- GWAS summary data by International Genomics of Alzheimer's Project (IGAP)
 - ~17K cases and ~37K controls of European ancestry

PrediXcan vs. DPR

- 5-fold cross validation
- Compare average prediction R^2
- Use threshold of prediction $R^2 > 0.01$ to identify significant gene expression prediction models



Manhattan Plots of TWAS results by S-PrediXcan Z-score Test Statistic



Omnibus-TWAS (TWAS-O, Hu et. al., AJHG 2024)

Combine TWAS Results of PrediXcan/FUSION/TIGAR by Aggregated Cauchy Association Test (ACAT)

- p_k : TWAS p-value using the k^{th} TWAS tool:

$$T_{ACAT} = \frac{1}{K} \sum_{k=1}^K \tan\{(0.5 - p_k) \pi\}$$

An average of Cauchy variables is approximated by a Cauchy distribution

$$p_{ACAT} \approx \frac{1}{2} - \frac{\{\arctan(T_{ACAT})\}}{\pi}$$

Cauchy transformed p-values:
Cauchy distributed under null hypothesis

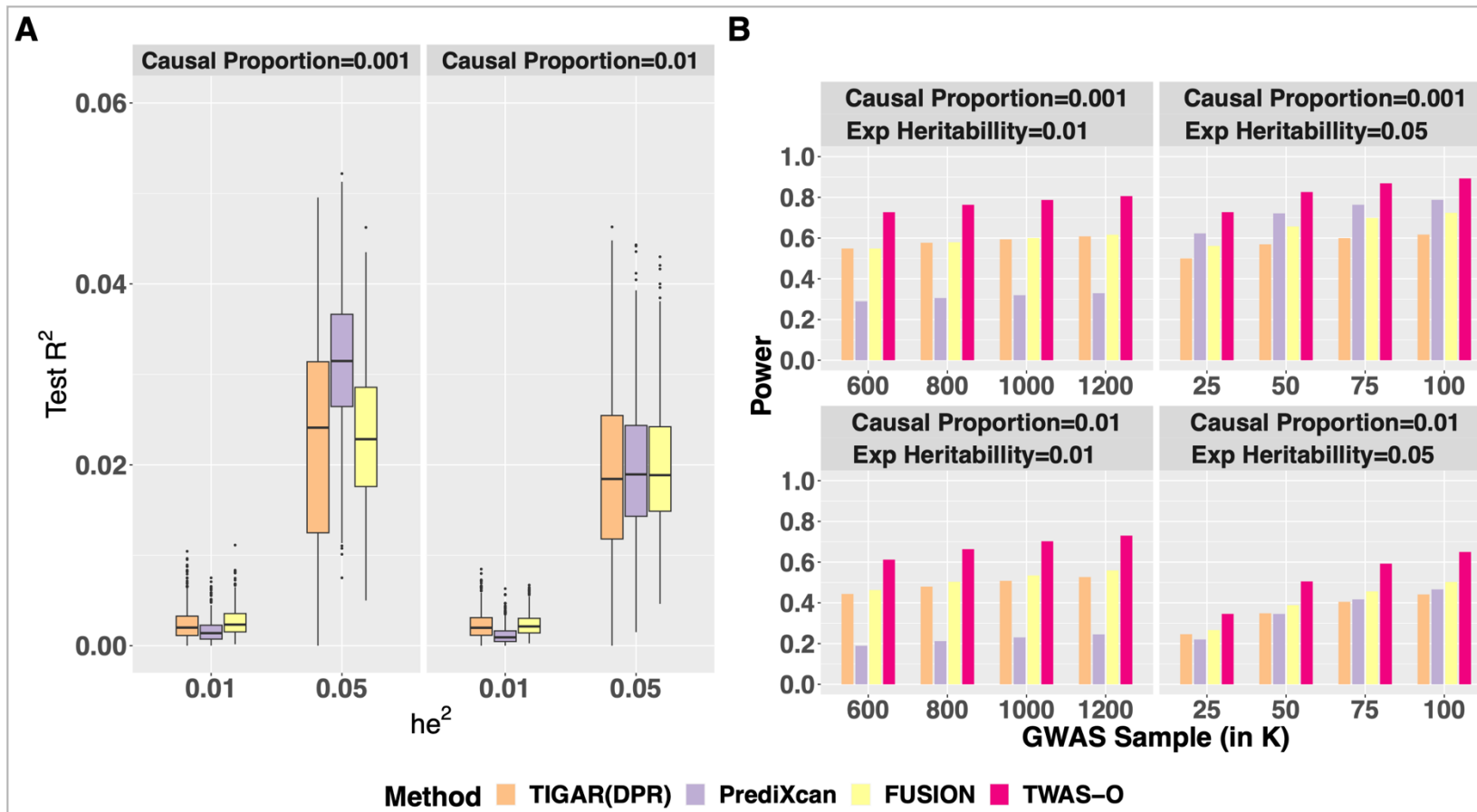
- Transformed p values increase quickly as the p value approaches 0, T_{ACAT} is essentially dominated by the components with very small p values;
- **If one of the TWAS tools give very small p-value, p_{ACAT} will also be very small.**
- Thus, ACAT can combine the strength of multiple TWAS tools and perform an omnibus test.

Simulations

- 1894 WGS European subjects
 - Randomly select 30% for training and 70% for prediction to mimic real training sample size
- Genotypes: randomly select 500 genes with various gene length and their cis-SNPs
 - Randomly select causal eQTLs: $p_{causal} = (0.001, 0.01)$
 - Proportion of expression variance explained by causal eQTLs: $h_e^2 = (0.01, 0.05)$
- Generated GWAS Z-scores: $Z \sim MVN(\Sigma_g \mathbf{w} \sqrt{n_{gwas} * h_{pheno}^2}, \Sigma_g)$
 - $h_{pheno}^2 = 0.025$: Proportion of phenotypic variance explained by simulated expression ($X_g \mathbf{w}$)
 - n_{gwas} : GWAS sample size
 - Σ_g : Correlation matrix of standardized genotype matrix X_g

Simulation Studies Showed higher power for the Omnibus TWAS Approach

Hu et. al., AJHG 2024



Increased Sample Sizes for Both Training and Test Cohorts

- Training transcriptomic reference data profiled by RNAseq:
 - ~1200 samples of dorsolateral prefrontal cortex from ROS/MAP cohorts
 - European ancestry
- Training genotype data profiled by Whole Genome Sequencing
- Test GWAS summary data by Bellenguez C. et al, Nat. Genetics, 2022
 - 39,106 European ancestry clinically diagnosed cases
 - 46,828 European ancestry proxy cases
 - 401,577 European ancestry controls
 - 75 GWAS Loci were identified

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New insights into the genetic etiology of Alzheimer's disease and related dementias

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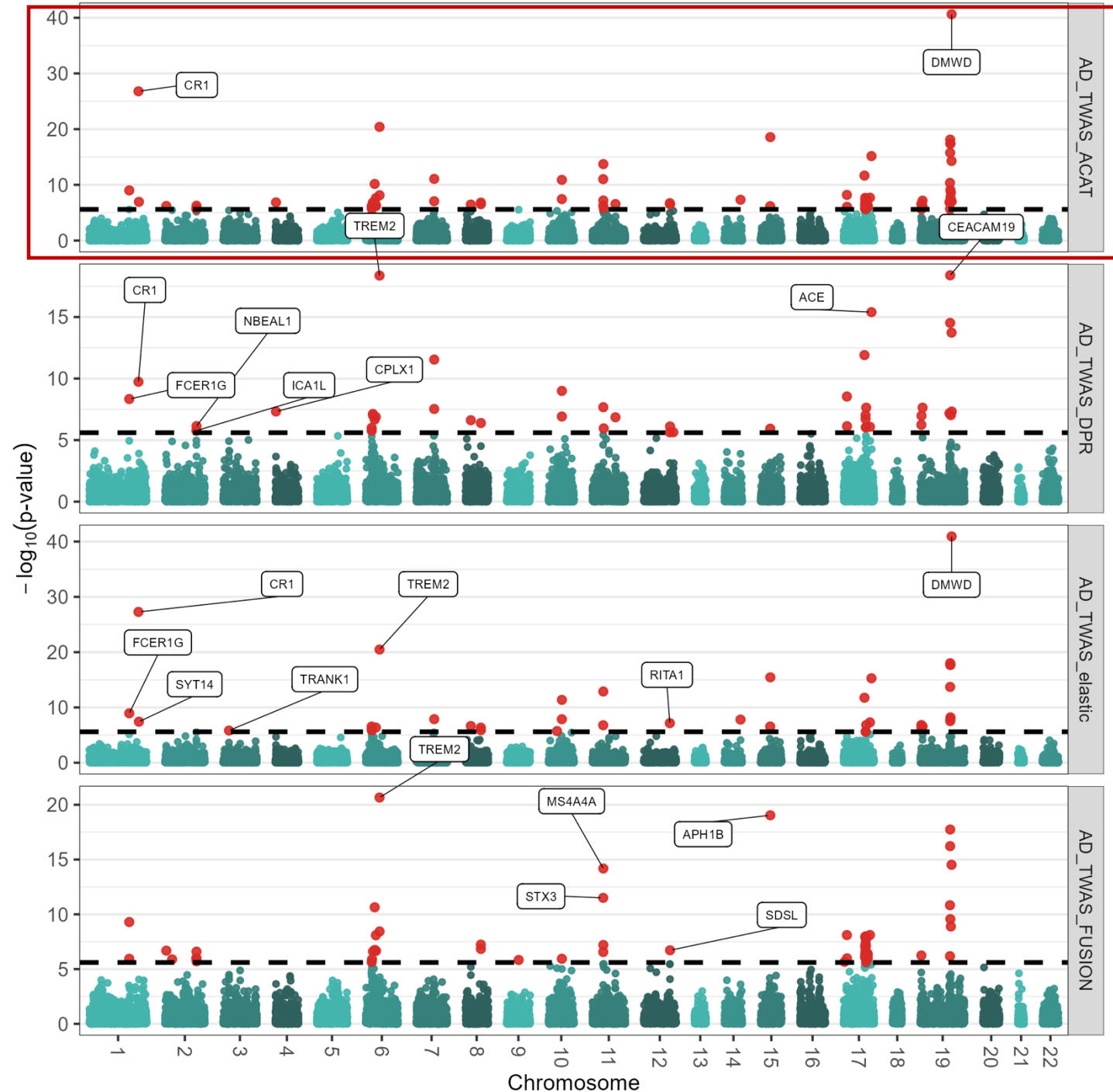
[Nature Genetics](#) 54, 412–436 (2022) | [Cite this article](#)

TWAS-O results of AD Dementia using ROSMAP bulk RNAseq data (n=~1200) of DLPFC and Bellenguez et. al. 2022 GWAS Summary data



Qiang (Leo) Liu

T. Hu, Q. Liu, et. al.
medRxiv [Preprint].
2024 Sep. PMID:
38585769



Web Resources

- GWAS Tool PLINK:
 - <https://www.cog-genomics.org/plink/2.0/>
- tensorQTL:
<https://github.com/broadinstitute/tensorqtl>
- FUSION
 - <http://gusevlab.org/projects/fusion/>
- TIGAR/DPR/EN
 - <https://github.com/yanglab-emory/TIGAR>
- Omnibus TWAS/PWAS Pipeline by ACAT
 - <https://github.com/tingyhu45/PWAS-O>